ScholarTrack API Documentation

Version: 1.0.0
Base URL: `http://localhost:5000/api`
Authentication: JWT Bearer Token (`Authorization: Bearer <token>`)
Data Format: JSON

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1. General Information

1.1 Authentication

Authenticated endpoints require a JWT bearer token obtained via `POST /api/auth/login` . Include the token in the `Authorization` header:

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIs...
```

Tokens expire after **24 hours**.

1.2 Pagination

List endpoints support pagination via query parameters:

Parameter	Type	Default	Description
`page`	integer	1	Page number
`per_page`	integer	20	Items per page (max 100)

Paginated responses include:

```
{
  "data": [ ... ],
  "pagination": {
    "total": 100,
    "page": 1,
    "per_page": 20,
    "pages": 5
  },
  "_links": {
    "self": "/api/...?page=1",
    "next": "/api/...?page=2",
    "prev": null
  }
}
```

1.3 Error Format

All errors follow a consistent format:

```
{
  "error": "not_found",
  "message": "Paper with arxiv_id '2403.99999' not found"
}
```

1.4 HTTP Status Codes

Code	Meaning	Usage
200	OK	Successful read or update
201	Created	Resource created successfully
204	No Content	Resource deleted successfully

Code	Meaning	Usage
400	Bad Request	Input validation failed
401	Unauthorized	Missing or invalid JWT token
403	Forbidden	Access denied to another user's resource
404	Not Found	Resource does not exist
409	Conflict	Duplicate operation (e.g. saving a paper twice)
500	Internal Server Error	Unexpected server error

1.5 HATEOAS Links

All responses include a `_links` object with related resource URIs, enabling API navigation without hardcoded paths.

2. F1: Authentication

2.1 POST /api/auth/register

Register a new user account.

Request Body:

Field	Type	Required	Description
`username`	string	Yes	Unique username
`email`	string	Yes	Valid email address
`password`	string	Yes	Account password

Example Request:

```
{
  "username": "researcher1",
  "email": "researcher1@example.com",
  "password": "securepass123"
}
```

Success Response (201 Created):

```
{
  "data": {
    "id": 1,
    "username": "researcher1",
    "email": "researcher1@example.com",
    "preferred_categories": [],
    "created_at": "2026-04-03T10:00:00"
  },
  "_links": {
    "self": "/api/users/me",
    "login": "/api/auth/login"
  }
}
```

Error Responses:

Code	Condition
400	Missing required fields or invalid email format
409	Username or email already exists

2.2 POST /api/auth/login

Authenticate a user and obtain a JWT access token.

Request Body:

Field	Type	Required	Description
`username`	string	Yes	Account username

Field	Type	Required	Description
`password`	string	Yes	Account password

Example Request:

```
{
  "username": "researcher1",
  "password": "securepass123"
}
```

Success Response (200 OK):

```
{
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIs...",
    "token_type": "bearer",
    "expires_in": 86400
  }
}
```

Error Responses:

Code	Condition
401	Invalid username or password

2.3 GET /api/users/me

Retrieve the current user's profile. **Requires authentication.**

Success Response (200 OK):

```
{
  "data": {
    "id": 1,
    "username": "researcher1",
    "email": "researcher1@example.com",
    "preferred_categories": ["cs.CV", "cs.CL"],
    "created_at": "2026-04-03T10:00:00"
  },
  "_links": {
    "self": "/api/users/me",
    "library": "/api/library",
    "notes": "/api/notes"
  }
}
```

Error Responses:

Code	Condition
401	Missing or invalid JWT token

2.4 PUT /api/users/me

Update the current user's profile. **Requires authentication.**

Request Body (all fields optional):

Field	Type	Description
`email`	string	New email address
`preferred_categories`	array of strings	Preferred arXiv categories

Example Request:

```
{
  "email": "new_email@example.com",
  "preferred_categories": ["cs.CV", "cs.CL", "cs.AI"]
}
```

Success Response (200 OK): Returns the updated user profile (same format as GET).

Error Responses:

Code	Condition
400	Invalid email format or invalid field types
401	Missing or invalid JWT token
409	Email already in use by another account

3. F2: Paper Discovery

3.1 GET /api/papers/search

Search arXiv papers by keyword. Results are fetched from the arXiv API and cached locally.

Query Parameters:

Parameter	Type	Required	Description
`q`	string	Yes	Search keywords
`category`	string	No	arXiv category filter (e.g. `cs.CV`)
`date_from`	string	No	Start date (YYYY-MM-DD)
`date_to`	string	No	End date (YYYY-MM-DD)
`page`	integer	No	Page number (default 1)
`per_page`	integer	No	Items per page (default 20, max 100)

Example Request:

```
GET /api/papers/search?q=vision+transformer&category=cs.CV&page=1
```

Success Response (200 OK):

```
{
  "data": [
    {
      "arxiv_id": "2403.12345",
      "title": "Efficient Vision Transformers with Multi-Scale Attention",
      "authors": ["Alice Smith", "Bob Johnson"],
      "abstract": "We propose a novel multi-scale attention mechanism...",
      "categories": "cs.CV cs.AI",
      "published_date": "2026-03-28",
      "arxiv_url": "https://arxiv.org/abs/2403.12345",
      "pdf_url": "https://arxiv.org/pdf/2403.12345",
      "_links": {
        "self": "/api/papers/2403.12345",
        "save": "/api/papers/2403.12345/save",
        "notes": "/api/papers/2403.12345/notes"
      }
    }
  ],
  "pagination": {
    "total": 45,
    "page": 1,
    "per_page": 20,
    "pages": 3
  },
  "_links": {
    "self": "/api/papers/search?q=vision+transformer&page=1&per_page=20",
    "next": "/api/papers/search?q=vision+transformer&page=2&per_page=20"
  }
}
```


Error Responses:

Code	Condition
400	Missing `q` parameter

3.2 GET /api/papers/trending

Get recent papers in a specific arXiv category.

Query Parameters:

Parameter	Type	Required	Description
`category`	string	Yes	arXiv category (e.g. `cs.CV`)
`days`	integer	No	Lookback period in days (default 7)
`page`	integer	No	Page number (default 1)
`per_page`	integer	No	Items per page (default 20, max 100)

Example Request:

```
GET /api/papers/trending?category=cs.CV&days=7
```

Success Response (200 OK): Same format as paper search.

Error Responses:

Code	Condition
400	Missing `category` parameter

3.3 GET /api/papers/

Retrieve details for a single paper. Checks local cache first, then fetches from arXiv API.

Path Parameters:

Parameter	Type	Description
`arxiv_id`	string	arXiv paper ID (e.g. `2403.12345`)

Success Response (200 OK):

```
{
  "data": {
    "arxiv_id": "2403.12345",
    "title": "Efficient Vision Transformers with Multi-Scale Attention",
    "authors": ["Alice Smith", "Bob Johnson"],
    "abstract": "We propose a novel multi-scale attention mechanism...",
    "categories": "cs.CV cs.AI",
    "published_date": "2026-03-28",
    "arxiv_url": "https://arxiv.org/abs/2403.12345",
    "pdf_url": "https://arxiv.org/pdf/2403.12345",
    "is_saved": true,
    "_links": {
      "self": "/api/papers/2403.12345",
      "save": "/api/papers/2403.12345/save",
      "notes": "/api/papers/2403.12345/notes"
    }
  }
}
```

Note: The `is_saved` field is only included when the request includes a valid JWT token.

Error Responses:

Code	Condition
404	Paper not found on arXiv

4. F3: Paper Management

4.1 POST /api/papers/{arxiv_id}/save

Save a paper to the user's personal library. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`arxiv_id`	string	arXiv paper ID

Request Body (optional):

Field	Type	Required	Description
`memo`	string	No	Personal note about the paper

Example Request:

```
{
  "memo": "Interesting multi-scale attention approach, read later"
}
```

Success Response (201 Created):

```
{
  "data": {
    "arxiv_id": "2403.12345",
    "title": "Efficient Vision Transformers with Multi-Scale Attention",
    "memo": "Interesting multi-scale attention approach, read later",
    "tags": [],
    "saved_at": "2026-04-03T14:30:00"
  },
  "_links": {
    "self": "/api/library",
    "paper": "/api/papers/2403.12345",
    "notes": "/api/papers/2403.12345/notes",
    "tags": "/api/library/2403.12345/tags"
  }
}
```

Error Responses:

Code	Condition
401	Missing or invalid JWT token
404	Paper not found (locally or on arXiv)
409	Paper already saved to library

4.2 DELETE /api/papers/{arxiv_id}/save

Remove a paper from the user's library. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`arxiv_id`	string	arXiv paper ID

Success Response (204 No Content): Empty body.

Error Responses:

Code	Condition
401	Missing or invalid JWT token
404	Paper not in the user's library

4.3 GET /api/library

List all papers in the user's personal library. **Requires authentication.**

Query Parameters:

Parameter	Type	Required	Description
`tag`	string	No	Filter by tag name
`page`	integer	No	Page number (default 1)
`per_page`	integer	No	Items per page (default 20, max 100)

Example Request:

```
GET /api/library?tag=attention&page=1
```

Success Response (200 OK):

```
{
  "data": [
    {
      "arxiv_id": "2403.12345",
      "title": "Efficient Vision Transformers with Multi-Scale Attention",
      "authors": ["Alice Smith", "Bob Johnson"],
      "categories": "cs.CV cs.AI",
      "published_date": "2026-03-28",
      "memo": "Interesting multi-scale attention approach",
      "tags": ["attention", "to read"],
      "saved_at": "2026-04-03T14:30:00",
      "_links": {
        "paper": "/api/papers/2403.12345",
        "notes": "/api/papers/2403.12345/notes",
        "tags": "/api/library/2403.12345/tags"
      }
    }
  ],
  "pagination": {
    "total": 12,
    "page": 1,
    "per_page": 20,
    "pages": 1
  },
  "_links": {
    "self": "/api/library?page=1&per_page=20"
  }
}
```

Error Responses:

Code	Condition
401	Missing or invalid JWT token

4.4 POST /api/library/{arxiv_id}/tags

Add tags to a saved paper. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`arxiv_id`	string	arXiv paper ID

Request Body:

Field	Type	Required	Description
`tags`	array of strings	Yes	Tag names to add

Example Request:

```
{
  "tags": ["attention", "to read"]
}
```

Success Response (200 OK):

```
{
  "data": {
    "arxiv_id": "2403.12345",
    "tags": ["attention", "to read"]
  }
}
```

Error Responses:

Code	Condition
400	`tags` field is not a list
401	Missing or invalid JWT token
404	Paper not in the user's library

4.5 DELETE /api/library/{arxiv_id}/tags/

Remove a tag from a saved paper. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`arxiv_id`	string	arXiv paper ID
`tag_name`	string	Tag name to remove

Success Response (204 No Content): Empty body.

Error Responses:

Code	Condition
401	Missing or invalid JWT token
404	Tag not found or paper not in library

4.6 GET /api/tags

List all of the current user's tags with usage counts. **Requires authentication.**

Success Response (200 OK):

```
{
  "data": [
    { "name": "attention", "count": 5 },
    { "name": "to read", "count": 3 },
    { "name": "important", "count": 2 }
  ]
}
```

`count` is the number of papers tagged with this label.

Error Responses:

Code	Condition
401	Missing or invalid JWT token

5. F4: Notes

5.1 POST /api/notes

Create a research note, optionally linked to a paper. **Requires authentication.**

Request Body:

Field	Type	Required	Description
`title`	string	Yes	Note title
`content`	string	Yes	Note content (Markdown supported)
`paper_id`	string	No	arXiv ID to link the note to

Example Request:

```
{
  "paper_id": "2403.12345",
  "title": "Multi-Scale Attention Notes",
  "content": "## Key Ideas\n\n- Proposes a hierarchical attention mechanism..."
}
```

Success Response (201 Created):

```
{
  "data": {
    "id": 1,
    "paper_id": "2403.12345",
    "title": "Multi-Scale Attention Notes",
    "content": "## Key Ideas\n\n- Proposes a hierarchical attention mechanism...",
    "created_at": "2026-04-03T15:00:00",
    "updated_at": "2026-04-03T15:00:00",
    "_links": {
      "self": "/api/notes/1",
      "paper": "/api/papers/2403.12345",
      "flashcards": "/api/notes/1/flashcards"
    }
  }
}
```

Error Responses:

Code	Condition
400	Missing `title` or `content`
401	Missing or invalid JWT token
404	Paper with given `paper_id` not found

5.2 GET /api/notes

List all notes for the current user. **Requires authentication.**

Query Parameters:

Parameter	Type	Required	Description
`paper_id`	string	No	Filter by arXiv paper ID
`page`	integer	No	Page number (default 1)
`per_page`	integer	No	Items per page (default 20, max 100)

Success Response (200 OK):

```
{
  "data": [
    {
      "id": 1,
      "paper_id": "2403.12345",
      "paper_title": "Efficient Vision Transformers...",
      "title": "Multi-Scale Attention Notes",
      "content": "## Key Ideas\n\n- ...",
      "created_at": "2026-04-03T15:00:00",
      "updated_at": "2026-04-03T15:00:00",
      "_links": {
        "self": "/api/notes/1",
        "paper": "/api/papers/2403.12345",
        "flashcards": "/api/notes/1/flashcards"
      }
    }
  ],
  "pagination": { "total": 8, "page": 1, "per_page": 20, "pages": 1 },
  "_links": { "self": "/api/notes?page=1&per_page=20" }
}
```

Error Responses:

Code	Condition
401	Missing or invalid JWT token

5.3 GET /api/papers/{arxiv_id}/notes

List notes linked to a specific paper. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`arxiv_id`	string	arXiv paper ID

Success Response (200 OK): Same format as `GET /api/notes`, filtered by paper.

Error Responses:

Code	Condition
401	Missing or invalid JWT token
404	Paper not found

5.4 GET /api/notes/

Retrieve a single note. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Note ID

Success Response (200 OK):

```
{
  "data": {
    "id": 1,
    "paper_id": "2403.12345",
    "paper_title": "Efficient Vision Transformers...",
    "title": "Multi-Scale Attention Notes",
    "content": "## Key Ideas\n\n- Proposes a hierarchical attention mechanism...",
    "created_at": "2026-04-03T15:00:00",
    "updated_at": "2026-04-03T15:00:00",
    "_links": {
      "self": "/api/notes/1",
      "paper": "/api/papers/2403.12345",
      "flashcards": "/api/notes/1/flashcards"
    }
  }
}
```

Error Responses:

Code	Condition
401	Missing or invalid JWT token

Code	Condition
403	Note belongs to another user
404	Note not found

5.5 PUT /api/notes/

Update a note. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Note ID

Request Body (all fields optional):

Field	Type	Description
`title`	string	Updated title
`content`	string	Updated content

Example Request:

```
{
  "title": "Updated Title",
  "content": "Updated content with new insights..."
}
```

Success Response (200 OK): Returns the updated note (same format as GET).

Error Responses:

Code	Condition
400	Empty request body
401	Missing or invalid JWT token

Code	Condition
403	Note belongs to another user
404	Note not found

5.6 DELETE /api/notes/

Delete a note and its associated flashcards. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Note ID

Success Response (204 No Content): Empty body.

Note: Deleting a note also cascade-deletes all linked flashcards and removes the note's vector from ChromaDB.

Error Responses:

Code	Condition
401	Missing or invalid JWT token
403	Note belongs to another user
404	Note not found

6. F5: Flashcards & Spaced Repetition

6.1 POST /api/flashcards

Create a flashcard linked to a note. **Requires authentication.**

Request Body:

Field	Type	Required	Description
`note_id`	integer	Yes	ID of the parent note
`question`	string	Yes	Flashcard question
`answer`	string	Yes	Flashcard answer

Example Request:

```
{
  "note_id": 1,
  "question": "What is Multi-Head Attention?",
  "answer": "A mechanism that runs multiple attention functions in parallel, allowing the model
}
```

Success Response (201 Created):

```
{
  "data": {
    "id": 1,
    "note_id": 1,
    "question": "What is Multi-Head Attention?",
    "answer": "A mechanism that runs multiple attention functions in parallel...",
    "ease_factor": 2.5,
    "interval": 1,
    "repetitions": 0,
    "next_review_at": "2026-04-04T15:00:00",
    "created_at": "2026-04-03T15:00:00",
    "_links": {
      "self": "/api/flashcards/1",
      "note": "/api/notes/1",
      "review": "/api/flashcards/1/review"
    }
  }
}
```

Error Responses:

Code	Condition
400	Missing required fields

Code	Condition
401	Missing or invalid JWT token
403	Note does not belong to the current user
404	Note not found

6.2 GET /api/flashcards

List all flashcards for the current user. **Requires authentication.**

Query Parameters:

Parameter	Type	Required	Description
`note_id`	integer	No	Filter by note ID
`page`	integer	No	Page number (default 1)
`per_page`	integer	No	Items per page (default 20, max 100)

Success Response (200 OK): Paginated list of flashcards.

Error Responses:

Code	Condition
401	Missing or invalid JWT token

6.3 GET /api/notes/{id}/flashcards

List flashcards for a specific note. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Note ID

Success Response (200 OK): Paginated list of flashcards (same format as `GET /api/flashcards`).

Error Responses:

Code	Condition
401	Missing or invalid JWT token
403	Note does not belong to the current user
404	Note not found

6.4 PUT /api/flashcards/

Update a flashcard's question or answer. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Flashcard ID

Request Body (all fields optional):

Field	Type	Description
`question`	string	Updated question
`answer`	string	Updated answer

Example Request:

```
{
  "question": "What are the components of Multi-Head Attention?",
  "answer": "Query, Key, and Value projections run through multiple parallel attention heads."
}
```

Success Response (200 OK): Returns the updated flashcard.

Error Responses:

Code	Condition
400	Empty request body
401	Missing or invalid JWT token
403	Flashcard does not belong to the current user
404	Flashcard not found

6.5 DELETE /api/flashcards/

Delete a flashcard. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Flashcard ID

Success Response (204 No Content): Empty body.

Error Responses:

Code	Condition
401	Missing or invalid JWT token
403	Flashcard does not belong to the current user
404	Flashcard not found

6.6 GET /api/flashcards/due

Get flashcards due for review today. **Requires authentication.**

Returns all flashcards where `next_review_at` is in the past.

Success Response (200 OK):

```
{
  "data": [
    {
      "id": 1,
      "question": "What is Multi-Head Attention?",
      "answer": "A mechanism that...",
      "note_id": 1,
      "ease_factor": 2.5,
      "interval": 1,
      "repetitions": 0,
      "next_review_at": "2026-04-04T15:00:00",
      "_links": {
        "self": "/api/flashcards/1",
        "review": "/api/flashcards/1/review",
        "note": "/api/notes/1"
      }
    }
  ],
  "total_due": 5
}
```

Error Responses:

Code	Condition
401	Missing or invalid JWT token

6.7 POST /api/flashcards/{id}/review

Submit a review rating for a flashcard. The SM-2 spaced repetition algorithm updates the card's scheduling parameters. **Requires authentication.**

Path Parameters:

Parameter	Type	Description
`id`	integer	Flashcard ID

Request Body:

Field	Type	Required	Description
`rating`	integer	Yes	Score from 0 to 5 (0 = forgot completely, 5 = perfect recall)

SM-2 Rating Scale:

Rating	Meaning
0	Complete blackout
1	Incorrect, but recognised the answer
2	Incorrect, but the answer seemed easy to recall
3	Correct with significant difficulty
4	Correct with some hesitation
5	Perfect recall

Example Request:

```
{
  "rating": 4
}
```

Success Response (200 OK):

```
{
  "data": {
    "flashcard_id": 1,
    "rating": 4,
    "updated": {
      "ease_factor": 2.5,
      "interval": 6,
      "repetitions": 2,
      "next_review_at": "2026-04-09T15:00:00"
    },
    "reviewed_at": "2026-04-03T20:00:00"
  }
}
```

Error Responses:

Code	Condition
400	Rating is not an integer between 0 and 5
401	Missing or invalid JWT token
403	Flashcard does not belong to the current user
404	Flashcard not found

6.8 GET /api/review/stats

Get spaced repetition review statistics. **Requires authentication.**

Query Parameters:

Parameter	Type	Required	Description
`period`	string	No	Time period: `week` (default), `month` , or `all`

Success Response (200 OK):

```
{
  "data": {
    "total_cards": 25,
    "due_today": 5,
    "mastered": 12,
    "learning": 13,
    "reviews_in_period": 42,
    "average_rating": 3.8,
    "daily_reviews": [
      { "date": "2026-04-01", "count": 8, "avg_rating": 3.5 },
      { "date": "2026-04-02", "count": 12, "avg_rating": 4.0 },
      { "date": "2026-04-03", "count": 22, "avg_rating": 3.9 }
    ]
  }
}
```

- `mastered` : Cards with `ease_factor >= 2.5` and `interval >= 21` days.
- `learning` : All other cards (`total_cards - mastered`).

Error Responses:

Code	Condition
401	Missing or invalid JWT token

7. F6: Semantic Search

All search endpoints use ChromaDB for vector similarity search. Relevance scores range from 0 to 1 (higher = more relevant), calculated as `1 - cosine_distance`.

7.1 POST /api/search/papers

Semantic search over paper abstracts. **No authentication required.**

Request Body:

Field	Type	Required	Description
`query`	string	Yes	Natural language search query
`n_results`	integer	No	Number of results (default 10, max 50)

Example Request:

```
{
  "query": "transformer attention mechanism optimization",
  "n_results": 10
}
```

Success Response (200 OK):

```
{
  "data": [
    {
      "arxiv_id": "2403.12345",
      "title": "Efficient Vision Transformers with Multi-Scale Attention",
      "abstract": "We propose a novel...",
      "categories": "cs.CV cs.AI",
      "published_date": "2026-03-28",
      "relevance_score": 0.89,
      "type": "paper",
      "_links": {
        "self": "/api/papers/2403.12345",
        "save": "/api/papers/2403.12345/save"
      }
    }
  ],
  "query": "transformer attention mechanism optimization",
  "source": "paper_abstracts"
}
```

Error Responses:

Code	Condition
400	Missing or empty `query` field

7.2 POST /api/search/notes

Semantic search over the current user's notes. **Requires authentication.**

Notes are isolated by user — each user can only search their own notes.

Request Body:

Field	Type	Required	Description
`query`	string	Yes	Natural language search query
`n_results`	integer	No	Number of results (default 5, max 50)

Example Request:

```
{
  "query": "data augmentation techniques",
  "n_results": 5
}
```

Success Response (200 OK):

```
{
  "data": [
    {
      "id": 3,
      "title": "Data Augmentation Survey Notes",
      "content": "## Summary\n\n- ...",
      "paper_id": "2403.67890",
      "relevance_score": 0.92,
      "type": "note",
      "_links": {
        "self": "/api/notes/3",
        "paper": "/api/papers/2403.67890"
      }
    }
  ],
  "query": "data augmentation techniques",
  "source": "user_notes"
}
```

Error Responses:

Code	Condition
400	Missing or empty `query` field
401	Missing or invalid JWT token

7.3 POST /api/search/all

Combined semantic search across both papers and notes, merged and sorted by relevance. **Requires authentication.**

Request Body:

Field	Type	Required	Description
`query`	string	Yes	Natural language search query
`n_results`	integer	No	Results per source (default 10, max 50)

Example Request:

```
{
  "query": "attention mechanism improvements",
  "n_results": 10
}
```

Success Response (200 OK):

```
{
  "data": [
    {
      "type": "note",
      "id": 1,
      "title": "Multi-Scale Attention Notes",
      "content": "Proposes a hierarchical attention mechanism...",
      "relevance_score": 0.94,
      "_links": { "self": "/api/notes/1" }
    },
    {
      "type": "paper",
      "arxiv_id": "2403.12345",
      "title": "Efficient Vision Transformers with Multi-Scale Attention",
      "abstract": "We propose a novel multi-scale attention...",
      "relevance_score": 0.89,
      "_links": { "self": "/api/papers/2403.12345" }
    }
  ],
  "query": "attention mechanism improvements",
  "source": "all"
}
```

Results are sorted by `relevance_score` in descending order. The `type` field indicates whether each result is a `paper` or `note`.

Error Responses:

Code	Condition
400	Missing or empty `query` field
401	Missing or invalid JWT token

8. Endpoint Summary

Module	Method	Path	Auth
F1	POST	`/api/auth/register`	No
	POST	`/api/auth/login`	No
	GET	`/api/users/me`	Yes
	PUT	`/api/users/me`	Yes
F2	GET	`/api/papers/search`	No
	GET	`/api/papers/trending`	No
	GET	`/api/papers/{arxiv_id}`	Optional
F3	POST	`/api/papers/{arxiv_id}/save`	Yes
	DELETE	`/api/papers/{arxiv_id}/save`	Yes
	GET	`/api/library`	Yes
	POST	`/api/library/{arxiv_id}/tags`	Yes
	DELETE	`/api/library/{arxiv_id}/tags/{tag_name}`	Yes
	GET	`/api/tags`	Yes
F4	POST	`/api/notes`	Yes
	GET	`/api/notes`	Yes
	GET	`/api/papers/{arxiv_id}/notes`	Yes
	GET	`/api/notes/{id}`	Yes
	PUT	`/api/notes/{id}`	Yes

Module	Method	Path	Auth
	DELETE	`/api/notes/{id}`	Yes
F5	POST	`/api/flashcards`	Yes
	GET	`/api/flashcards`	Yes
	GET	`/api/notes/{id}/flashcards`	Yes
	PUT	`/api/flashcards/{id}`	Yes
	DELETE	`/api/flashcards/{id}`	Yes
	GET	`/api/flashcards/due`	Yes
	POST	`/api/flashcards/{id}/review`	Yes
	GET	`/api/review/stats`	Yes
F6	POST	`/api/search/papers`	No
	POST	`/api/search/notes`	Yes
	POST	`/api/search/all`	Yes

Total: 30 endpoints (7 public + 1 optional auth + 22 authenticated)